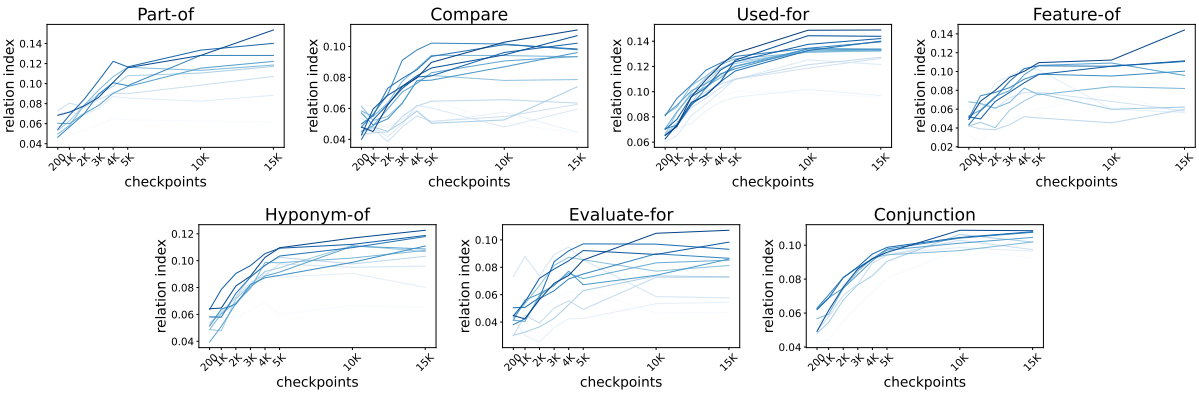

Figure 18: The change curve of relation indexes of attention heads for semantic relationships.

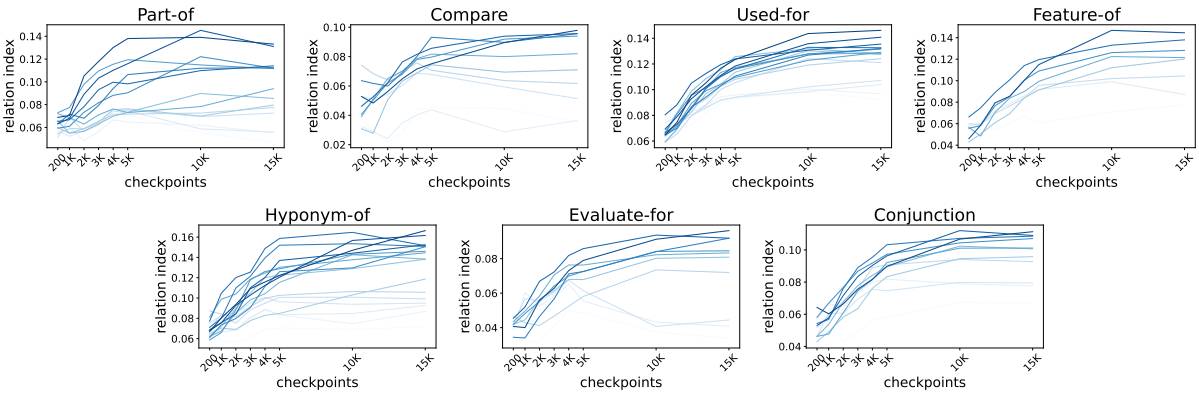

Figure 19: The change curve of relation indexes of attention heads for reverse semantic relationships.

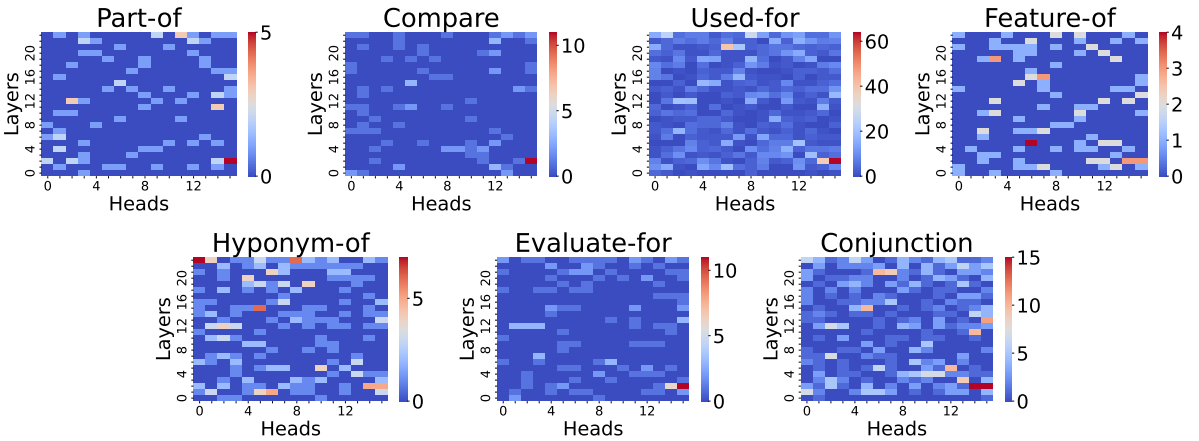

Figure 20: Heatmaps of occurrence of attention heads having the largest value of $\mathbb{E}_j[a_T^{h,j}]$ for each triplet in semantic relations.

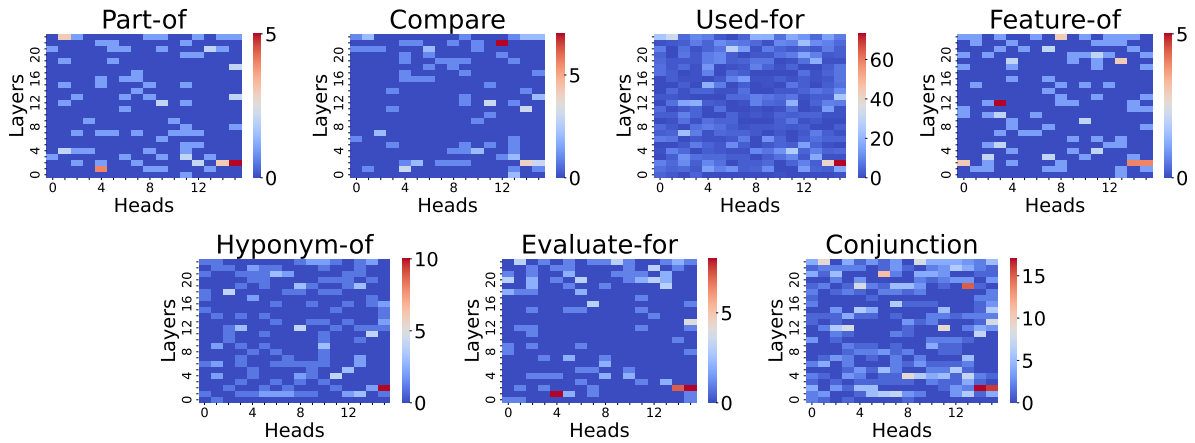

Figure 21: Heatmaps of occurrence of attention heads having the largest value of $\mathbb{E}_j[a_T^{h,j}]$ for each triplet in reverse semantic relations.